

GR&R; Study Report

Gauge Repeatability & Reproducibility Analysis

Study Date:

N/A

Equipment ID:

CMM-001

Characteristic:

diameter

Method:

N/A

Operators:

N/A

Parts:

N/A

Trials:

N/A

Generated By:

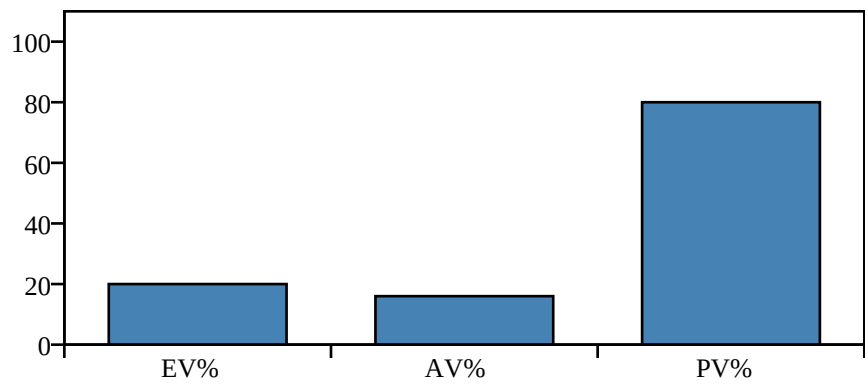
N/A

Component	Value	% Contribution
Repeatability (EV)	0.1000	20.0%
Reproducibility (AV)	0.0800	16.0%
Part Variation (PV)	0.4000	80.0%
Total Variation (TV)	0.5000	100%
GR&R %	18.50%	—
NDC	4	—

CONDITIONAL

%GR&R; = 18.5% — conditional
Engineering review required before production use.
Consider investigating repeatability as the dominant source.
NDC = 4 — below minimum of 5. Cannot distinguish part variation.
EV (Repeatability) = 0.1000
AV (Reproducibility) = 0.0800

Variance Component Breakdown



AIAG MSA Reference Guidelines

Threshold	Verdict	Guideline
< 10%	Acceptable	Suitable for production
10-30%	Conditional	Engineering review required
> 30%	Not Acceptable	Must improve before use
NDC ≥ 5	Adequate	Can distinguish parts